Discounted Permutation Relevance score as ranking performance quality evaluation metric

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Abstract

Ranking is a common output for many data-driven decision systems, in particular in the fields such as recommendation systems, information retrieval (IR), and attribution modeling. We believe that validation of the ranking solutions remains challenging due to a lack of good quality metrics that can provide both discounted graded relevance across all items of the rank as well as capture absolute differences even within the same ranks. The standard approach in the field is to use Normalized DCG (NDCG), but as it is using logarithmic discounting, only a few top-rank elements noticeably contribute to the final score. In this paper, we propose a pair of novel metrics that are superior in capturing differences in rankings. Our intention is to incorporate a higher importance of relative order even at the tail of the ranking. We evaluated our metrics against NDCG and demonstrated improved sensitivity to differences in the rankings.

1 Introduction

Motivation, background, and main contributions.

The necessity for a new metric arose during our work on multichannel marketing attribution solutions. We realized that traditional ranking quality metrics, which focus predominantly on the top of the list, fail to satisfy our needs—especially when contributions to the score decay exponentially. In our particular scenario, it proved problematic that certain marketing channels or touchpoints could be ignored and lost in the exponential tail of a ranking metric.

However, this limitation extends far beyond marketing. We believe that the proposed metric addresses a critical gap in a broad class of high-recall retrieval problems. In these domains, the user intent is not merely to find the best answer quickly, but rather to exhaustively surface all relevant information, making the evaluation of the entire ranked list critical. Prominent examples include:

- Legal e-Discovery: During litigation, legal counsel must surface every relevant communication or contract. Overlooking critical evidence simply because a search algorithm ranked it poorly can compromise a case.
- Patent Search (Prior Art): Before approving a new patent, examiners must verify that the invention does not already exist. If prior art is retrieved but buried at the bottom of a search list, it still legally invalidates the new patent.
- Systematic Medical Reviews: When researchers conduct meta-analyses (e.g., evaluating drug efficacy), they must aggregate every published study on the topic. Missing studies at the bottom of a ranking can alter the final medical consensus.
- Enterprise Compliance Auditing: Internal database searches for regulatory violations must return all instances of non-compliance, not just the most prominent ones.
- Plagiarism Copyright Enforcement: Systems identifying unauthorized use of intellectual property must flag every instance across millions of records, regardless of its rank order.

Therefore, in this paper, we will demonstrate the potential benefits of our new approach across these varied challenges. We highlight the missing gap in current evaluation methods and detail our implementation to provide a clear understanding of the metric's applicability.

2 Related work

Before we proceed into exploring relevant research activities and existing solutions in the field which altogether are going to help us to shape the scope of current work we would like to explicitly define the location by using existing categorizations. First we will explain categorizations themselves and after that explain where a focus of the paper is.

2.0.1 Offline vs Online evaluation metric

Offline evaluation metrics are the ones that are not being used as a target for optimization within your machine learning algorithm. It is either technically challenging due to the non-differentiable nature of the metric or it depends on the approach that is being used for learning the ranking. We will discuss it in the following section.

Online evaluation metrics on the other hand can be used during optimization process and of course of a greatest importance as they enable us to search for optimal solution algorithmically

2.0.2 Approaches to Learning to Rank

Algorithms can be grouped into three approaches: the pointwise approach, the pairwise approach, and the listwise approach [Liu T 2009 p240]

The pointwise approach The input space of the pointwise approach contains the feature vector of each single document. The pairwise approach The input space of the pairwise approach contains a pair of documents, both represented as feature vectors. The listwise approach The input space of the listwise approach contains the entire group of documents associated with a query.

2.0.3 Query-dependent vs query-independent models

These models explicitly use features derived from the interaction between the query and each document (e.g., text similarity, BM25 scores, query term matches) to evaluate relevance. Typical use is within search engines' core relevance ranking and learning to rank with query-document pairs.

(Query-independent models) These models rank documents based on their own importance. PageRank can be taken as an example of such an approach. PageRank is used as an independent, query-agnostic measure of a page's importance, and this score can be combined with query relevance signals when ranking search results. This means a page with a higher PageRank may still appear lower than a more relevant page for a specific search query.

Although some metrics can be usually used in for example query-dependent setups it doesn't mean that we can not use them for query-independent models when it happens that we know ground truth of the ranking and want to evaluate and compare different algorithms. This is one of the use cases that we found important while working on an evaluation of multichannel attribution modeling approaches. To give here a bit more context let's say we have a syntheticly generated dataset and we know which touchpoints in a customer journey should be seen as more important ones. Then we can apply multiple different modeling technics in order to find the true ranking. In this setup it is possible to use query-dependent metrics even though there is no notion of a query. We could evaluate performance of the PageRank, for example, if we knew in advance what should be a perfect ranking given the assumptions of the approach itself.

2.0.4 Graded vs binary relevance

Many of the ranking metrics consider only a binary relevance meaning they evaluate whether a certain item appeared in the rank or not.

- Precision metric easures the proportion of retrieved items in the top-k results that are relevant.
- AP is the average of precision values computed each time a relevant item is retrieved in the ranked list. MAP is the mean of AP over multiple queries.

- Mean Reciprocal Rank (MRR) measures the average of the reciprocal of the rank at which the first relevant item appears across all queries. Only the first relevant (binary) item in the list matters; all other ranks are ignored, and there is no notion of "degree" of relevance.

  Graded metrics on the other hand are designed to incorporate no-binary relevance.

- DCG/NDCG sums up the graded relevance scores of documents in the ranked list, penalizing relevant items that appear lower.

- Cumulative Gain (CG) simply the sum of graded relevance scores of the top-k items.Direct sum, does not account for rank/order—only total amount of relevance. link

- Expected Reciprocal Rank (ERR) extends the reciprocal rank idea to graded judgments, modeling the chance that a user will stop at a highly relevant document.

2.0.5 Focus of the current proposal

Given all the categorizations mentioned above we now can try to narrow down the scope of our work. This work is going to focus on both offline graded evaluation of query-independent models and online graded evaluation for listwise approaches in the learning-to-rank domain.

2.1 Feature attribution in explainable AI for ranking systems

2.1.1 Overview of the related work

The probably closest work to ours is [HdA24] that is touching the topic of importance of relative ranking evaluation but it explicitly focuses on a specific area of AI explainability and attributions of the features. Their work is motivated by the fact that feature attributions (read ranking) should take into account the way that ordered list was being built with. This paper is trying to tackle a more general problem and not particularly about understanding of a connection between features (input data), prediction methods and the relationships between elements of the ranked list.

What is still common between present work and [HdA24] article is an understanding of the lack of appropriate metrics that can capture relative position changes in the ranking across the whole list. So to say it in other words, even though motivation of our efforts are different we still share the same vision that available metrics for ranking are not covering all recent ranking evaluation use cases. In the listwise feature attribution work they are looking for a single value/score that will reflect impact of changes in input features to the changes/permutations in the output ranking. This work's goal, on the other hand, is to introduce a metric that can help to compare two versions of the ranking in a more general sense. We believe that this goal goes beyond a task of feature attribution and can be applied to a broader set of tasks such as recommendation system's ranking evaluation, marketing attribution validation, similarity based on preference lists etc.

It is worth to note that introduction of such a metric would also create another opportunity to improve feature attribution evaluation for ranking systems such as in [HdA24] as it will enable better comparison of two versions of the rankings: before and after changes that took place in the input data and feature space.

The following sections will provide some extra background that will be relevant for understanding present paper's direction and methods. This background does not necessarily mean that there is an active research in those fields.

2.2 Normalized Discounted Cumulative Gain metric

NDCG is a widely used metric for evaluating the quality of ranked lists, especially in information retrieval, search engines, and recommender systems.

The new metric that we are going to propose in this paper falls into the same category of ranking quality metrics as NDCG and will be heavily compared against it in an experimentation section.

2.2.1 When it is being used

NDCG is favored in contexts where the relative position of relevant items in a ranked list is crucial and where graded (non-binary) relevance is needed.

It is used both as an offline evaluation metric and, in some advanced systems, as a direct optimization target (using differentiable approximations or surrogate losses)

2.2.2 Important caveats

While many ranking algorithms can be evaluated with NDCG, optimizing for NDCG directly during training is technically challenging due to the non-differentiable and list-based nature of the metric. Hence, some algorithms approximate or surrogate the metric during model training

2.3 Kendall rank correlation coefficient

2.3.1 Definition

Kendall rank correlation coefficient [Ken38] (Kendall's Tau) is computed by comparing all possible pairs of observations to determine whether their orderings are concordant (agreeing) or discordant (disagreeing) when ranked based on two different variables.

2.3.2 Why is it relevant to our work

We are going to use Kendall's Tau's principle as part of our solution and that is why a brief overview might help in building overall understanding of a proposal. It will also be relevant when we will be evaluating strengths and weaknesses of the new approach.

2.3.3 Historical background

Kendall's Tau was first introduced by Maurice Kendall in 1938 as a method to measure ordinal association between two measured quantities. Since its introduction, it has become a foundational tool in statistics and has seen renewed popularity with the rise of data science and machine learning, thanks to modern computational optimizations.

2.3.4 Mathematical formulation

Kendall's Tau is computed by comparing all possible pairs of observations to determine whether their orderings are concordant (agreeing) or discordant (disagreeing) between two variables. The basic formula is:

$$\tau = \frac{C - D}{C + D}$$

where C is a number of concordant pairs and D is a number of discordant pairs

2.3.5 Boundaries

Another shared feature is the boundaries. Proposed metric will be positive unbounded number

2.4 Spearman's footrule distance

2.4.1 Definition

Spearman's footrule distance is a non-parametric metric that quantifies the difference between two ranked lists (permutations) by summing the absolute differences in the positions of each item. It is widely used in data science for:

2.4.2 Historical background

Spearman's footrule distance [DG77] was first introduced by Charles Spearman, a British psychologist, as an alternative to correlation for pairs of ranks. Its original use was in the context of measuring agreement between two rankings, and it has since found applications in statistics, computer science, and information retrieval. The concept dates back to the early 20th century, with renewed interest in the late 20th and 21st centuries due to its intuitive interpretation and computational simplicity.

2.4.3 Mathematical formulation

Given two rankings R_1 and R_2 of n items, the Spearman's footrule distance F is:

$$F(R_1, R_2) = \sum_{i=1}^{n} |R_1(i) - R_2(i)|$$

where $R_1(i)$ and $R_2(i)$ are the positions of item i in the two rankings.

3 Proposed metric - Discounted Permutation Relevance score

3.1 The proposal

With this work we would like to introduce a novel metric that we believe will enrich an existing toolkit for evaluation of ranking systems. We propose the name Discounted Permutation Relevance score as it captures major properties of the metric. DPR score ensures a linearly distributed decay of importance of changes across the whole ranking list meaning that any kind of discrepancies and changes at the top of the rank will contribute more to the final score. Term Permutation highlights the fact that approach is using permutations as one of the backbone mechanisms for evaluation of the information loss. DPR score is also taking into account relative pairwise changes within the rank hence the term relevance.

3.2 Mathematical formulation

Formulation is quite complex as it includes multiple aspects from other existing metrics as well as its own novelty approaches. We present a pseudocode description in the following sections instead.

3.3 What is being penalized

Direction of the shift on its own is not what is being penalized. Algorithm only marks areas where movements are more expensive.

3.4 Theoretical properties, bounds and limitations

DPR score has a linear decay of importance of changes so we should not expect overfitting to only a small part of the ranking.

Due to the fact that approach relies on permutations and number of permutations depends on size of the ranking list score doesn't have and upper bound.

Since this value reflects a difference, or equivalently a distance, between rankings, it is always non-negative. Interpreting it as a positive quantity makes reasoning about distances more straightforward.

Current version of the metric only considers cases when rankings are of the same size and both lists share the same set of elements.

Score could have been normalized to [0,1] range but that requires knowing of the furthest version of the ranking that would represent the worst score. Given the complexity of the algorithm this search of the worst case doesn't have an optimal analytical solution and all the stochastic solutions wouldn't guarantee that the found permutation of the list is the furthest in terms of a score. As a result normalization would increase computational complexity and cost of calculation so we decided to keep the metric as lightweight as possible.

3.5 Conceptual comparison with existing metrics

3.5.1 Relevance to the Kendall's Tau

Kendall's Tau uses idea of permutations as a way to capture all possible first order relationships within the ranking. In this paper, the approach is advanced further by integrating several additional properties into each permutation.

3.5.2 Relevance to the Spearman's footrule distance

Spearman's footrule distance is only comparing N pairs, where N is the size of the ranking list, taking into account only every label.

With DPR score we are computing absolute distance not only for every label but also for every combination involving that label. Scoring incorporates how element changes not only with respect to its previous position, but also with respect to all the other elements in the ranked list.

3.5.3 Relevance to the NDCG

Similar to NDCG metric our score takes into account position of the elements. Proximity to the top affects assigned importance and therefore enables discounted calculation. That also incorporates behavioral modal that assumes that users ore more likely to interact with (or be satisfied by) results near the top, In other words, DPR supports graded relevance as all the items in the ground truth list are considered to be sorted in descending order based on their relevance.

4 Implementation details

5 Experimental Setup

5.1 Datasets used (synthetic, real-world, benchmarks)

We used only synthetic datasets as our goal is to highlight specific properties of the metric and not to beat other metrics on a public benchmarks.

5.2 Tools used (programming language, libraries)

We used Python programming language and public python packages to perform experiment and comparisons.

6 Experiments

6.1 Comparing global sensitivity of NDCG and DPR score

6.1.1 Setup

We have two identical lists - one will be used as a ground truth ranking and a second one will be an identical copy of the ground truth list that will be gradually permuted through the whole space of permutations starting from the tail, one permutation level at a time. We will use the term *permutation level*, which we believe is not the common term in combinatorics, just for the simplicity sake given the context of the task. Otherwise this is more commonly described as recursion depth, subproblem size, or active subset.

6.1.2 Goal of the experiment

We wanted to show how sensitivity of the metrics behave as we keep permuting the ground truth list starting from the tail side.

First goal is to confirm that the behavior of the DPR score aligns with the proposed design - sensitivity changes more linearly and disagreements at the very tail of the ranking are being respected. Second goal, adjacent to the first one, is to show how limited sensitivity of the NDCG metric is.

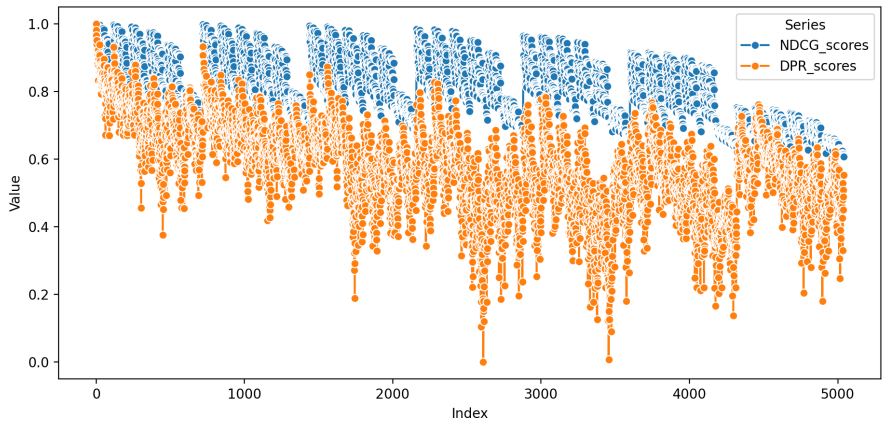

Figure 1: Global sensitivity analysis

6.1.3 Observations from the experiment

As can be seen in Figure 1 within each permutation level we see less variance in value for NDCG metric. It reflects the fact that metric is highly dependent on mostly one thing - distance from the top. DPR on the other hand tries to capture other notions of disagreement and therefore may have more similarly penalized versions of the ranking when different notions compete for major penalty contribution.

Pattern for the NDCG is different and shows that up until the very top of the ranking (top 3 permutation levels) sensitivity is limited.

6.2 Comparing local sensitivity of NDCG and DPR score

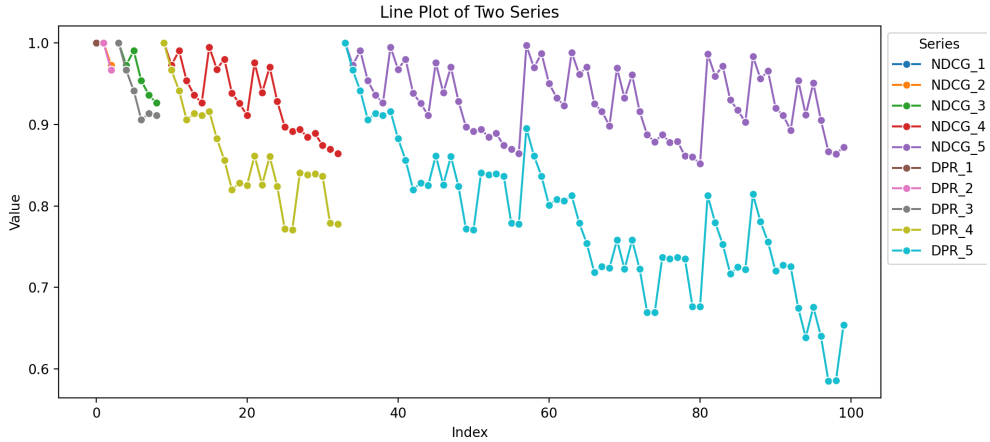

Figure 2: Local sensitivity analysis

6.3 Highlighting limitations of the NDCG

In order to understand better differences between DPR and NDCG we run through all possible list's permutations in a hope to find interesting behaviour. Below are some of those examples:

6.3.1 Example 1

```
[4, 0, 3, 2, 5, 1, 6] -> [5, 0, 3, 2, 4, 1, 6]
NDCG 0.8939 -> 0.9325
DPR  0.6678 -> 0.6355
```

6.3.2 Example 2

```
[4, 6, 2, 0, 5, 3, 1] -> [4, 6, 0, 2, 5, 3, 1]
NDCG 0.7102 -> 0.7062
DPR  0.4989 -> 0.5224
```

4 and 5 swap. In the ground truth ranking 4 and 5 have distance 1 (next to each other, difference between indexes is 1). Then if the order of this pair is changed in a way that originally less relevant element took place of a more relevant one we should expect that the agreement with the original ranking should decrease. But according to the NDCG score this transformation has brought the agreement down.

7 Results

Quantitative results (tables, graphs).

Qualitative analysis (case studies, examples).

Statistical significance (if applicable).

8 Discussions and Future work

- We are considering only pairwise permutations whereas relationships of higher order are possible and do exist. Let's say we could evaluate triplets of values and see how their internal relationships change with respect to the ground truth positions.

- This is a first version of the metric and we hope that we will learn more about its applicability, efficiency and ways to improve it as we receive feedback from the community and our own experience.

References

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